

# Architecture Selection Justification

**Architecture Selection:** "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

**Architectural Choice:** Layered Architecture

## Two Reasons:

- **Separation of Concerns:** It neatly decouples the user-facing touch screen interface (Presentation Layer) from the business logic (Order and Payment Management) and data storage (Menu/Pricing). This makes the system easier to maintain and update over time.
- **Hardware Isolation:** The architecture isolates hardware-specific integrations (like the touch screen and receipt printer) into specific layers. If a printer model changes, only the Receipt Printer Component in the Business Layer needs updating, without affecting the UI or database.

**Security Advantage:** The layered approach inherently restricts direct access from the Presentation Layer to the Data Layer. All interactions must flow through the Business Layer (Order Manager and Payment Service). This allows the Payment Service Component to securely process credit card transactions and validate inputs before any data is recorded, minimizing the risk of unauthorized database access or injection attacks.

**Performance Benefit:** Since the system handles a fixed set of options (3 coffee types, 2 sizes), the Business Layer can cache menu and pricing information retrieved from the Data Layer. This reduces the number of database queries required during order customization, leading to a faster and more responsive touch screen experience for customers in a busy cafe environment.